

3d Map Reconstruction

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1 Local Map Reconstruction

We first reconstruct a local map for each selected group of images. For each group, we estimate the relative camera poses from corresponding object corners, determine metric scale from sensor camera positions, and triangulate the object corners. We then refine the camera poses and reconstructed points using bundle adjustment (BA). The accepted local maps provide the input for global map reconstruction.

1.1 Preparing Correspondences and Selecting Anchors

First, we load the image groups, camera calibration, sensor poses, and manually annotated object correspondences. Each object has a consistent object ID and four ordered corners: top-left, top-right, bottom-right, and bottom-left.

For each group, we identify objects with fully visible and valid corner annotations in every frame. We remove lens distortion from their image coordinates before estimating the geometry.

Next, we select anchor objects distributed across the images to initialize the camera poses. We evaluate several spatially distributed subsets, together with a candidate containing all common objects. These anchors provide the initial geometry; they do not limit which objects can appear in the final local map.

1.2 Estimating Relative Camera Poses

For each image pair, we use the corresponding anchor corners to estimate an essential matrix with RANSAC. For undistorted normalized image coordinates, the epipolar constraint is

$$\tilde{\mathbf{p}}_b^T \mathbf{E}_{ba} \tilde{\mathbf{p}}_a = 0, \quad \tilde{\mathbf{p}} = \mathbf{K}^{-1} \begin{bmatrix} u & v & 1 \end{bmatrix}^T, \quad (1)$$

where $\mathbf{K}$ is the intrinsic matrix.

We decompose the essential matrix to obtain the relative rotation $\mathbf{R}_{ba}$ and translation direction $\hat{\mathbf{t}}_{ba}$:

$$\mathbf{E}_{ba} \propto [\hat{\mathbf{t}}_{ba}]_{\times} \mathbf{R}_{ba}. \quad (2)$$

Here, $[\hat{\mathbf{t}}_{ba}]_{\times}$ is the matrix representing the cross product with the translation direction. The relative pose maps coordinates from camera a into camera b .

The essential matrix provides the translation direction but not its metric length. We obtain the baseline from the sensor camera centres:

$$b_{ab} = \|\mathbf{C}_b^{\text{sensor}} - \mathbf{C}_a^{\text{sensor}}\|. \quad (3)$$

We then initialize the metric translation as $\mathbf{t}_{ba} = b_{ab}\hat{\mathbf{t}}_{ba}$.

When camera a is the reference, its pose is $\mathbf{R}_a = \mathbf{I}$ and $\mathbf{C}_a = \mathbf{0}$. The centre of camera b in this reference system is

$$\mathbf{C}_b = -\mathbf{R}_{ba}^{\top} \mathbf{t}_{ba}. \quad (4)$$

1.3 Checking Rotation Consistency and Initializing the Map

For three cameras, independently estimated relative rotations should approximately satisfy

$$\mathbf{R}_{ca} \approx \mathbf{R}_{cb} \mathbf{R}_{ba}. \quad (5)$$

We measure their disagreement as a rotation-cycle error. This error indicates inconsistency between the pair estimates, but it does not automatically reject the group before BA.

We first try to initialize all camera poses using their estimated relationships to one reference camera. If this initialization does not produce sufficient valid geometry, we initialize the map from a supported camera pair. We triangulate its anchor corners and use Perspective-n-Point (PnP) estimation to register the remaining cameras from their 2D observations of these initial 3D points.

This gives each registered camera one pose in a common local coordinate system, which we can refine jointly.

1.4 Triangulating and Refining the Anchor Geometry

Using the initial camera poses, we form the projection matrix

$$\mathbf{P}_f = \mathbf{K} [\mathbf{R}_f \quad -\mathbf{R}_f \mathbf{C}_f]. \quad (6)$$

We then triangulate each anchor corner from its corresponding observations in multiple images. The linear triangulation equations are solved using singular value decomposition, providing an initial 3D position for each corner.

Next, we apply BA to jointly refine the camera rotations, camera centres, and anchor corners while keeping the reference-camera pose fixed. The predicted image position of a corner is

$$\hat{\mathbf{p}}_{fj} = \pi(\mathbf{K} \mathbf{R}_f (\mathbf{X}_j - \mathbf{C}_f)), \quad \pi([x, y, z]^{\top}) = [x/z, y/z]^{\top}. \quad (7)$$

BA reduces the differences between the projected corners and their annotated, undistorted image coordinates. We use a robust loss for the pixel residuals and include sensor-derived camera-pair distances as metric constraints. The camera calibration remains fixed.

1.5 Reconstructing the Remaining Objects

After refining the anchor geometry, we revisit all common objects, including anchors that were not retained during initialization.

For each object, we first attempt reconstruction using all available views. If these observations are inconsistent, we test smaller view subsets. This allows an object to remain in the map even when one of its image observations is unsuitable.

We refine the triangulated corners and then perform joint BA using the accepted objects and camera poses. An image observation is retained only when all four corners lie in front of the camera and each corner’s reprojection error is at most 3 px. An accepted object must have at least two supporting views, an RMSE no greater than 2 px, and sufficient separation between its viewing rays.

We compare the final reconstructions obtained from the different anchor candidates. We prefer the candidate retaining the most objects, then the most corner observations, and finally the lowest reprojection RMSE. If no candidate passes the reconstruction checks, the group does not produce an accepted local map.

1.6 Object Measurements and World Coordinates

For each accepted object, we calculate its centre as the mean of its four reconstructed corners. We estimate width and height by averaging the corresponding opposite edge lengths:

$$w = \frac{\|\mathbf{X}_{TR} - \mathbf{X}_{TL}\| + \|\mathbf{X}_{BR} - \mathbf{X}_{BL}\|}{2}, \quad (8)$$

$$h = \frac{\|\mathbf{X}_{BL} - \mathbf{X}_{TL}\| + \|\mathbf{X}_{BR} - \mathbf{X}_{TR}\|}{2}. \quad (9)$$

We also transform the local coordinates into the world coordinate system using the reference camera’s sensor pose:

$$\mathbf{X}_{\text{world}} = \mathbf{Q}_{\text{ref}} \mathbf{X}_{\text{local}} + \mathbf{C}_{\text{ref}}^{\text{sensor}}, \quad (10)$$

where $\mathbf{Q}_{\text{ref}}$ is the reference camera’s camera-to-world rotation. This world placement is provisional, because it inherits the uncertainty of the reference sensor pose.

1.7 Example: Local Map G0001

We illustrate the reconstruction using group G0001, which contains the following four frames:

- 0051_POS_073,
- 0052_POS_105,
- 0053_POS_105,

- 0054_POS_113.

The group contains 22 fully visible common objects. The selected candidate, S01, uses seven anchor objects with ID suffixes 001, 024, 025, 026, 034, 042, and 046. Their four corners provide 28 input correspondences for each image pair.

The largest initial rotation-cycle error is 2.675° . Although this exceeds the configured diagnostic warning threshold of 2° , the group is allowed to continue to refinement.

The initialization directly from the reference camera did not retain sufficient connected support. The successful reconstruction instead started from the pair 0051_POS_073 and 0054_POS_113, followed by PnP registration of the remaining cameras and joint BA.

The final local map retains all 22 common objects: 17 glass objects, two frames, two windows, and one darkening object. Table 1 summarizes its geometry and reconstruction consistency.

Table 1: Geometry and reconstruction results for local map G0001.

Metric	Result
Camera frames	4
Common objects	22
Selected anchor objects	7
Reconstructed objects	22
Reconstructed 3D corners	88
Retained corner observations	324
Maximum initial rotation-cycle error	2.675°
RMSE before the last BA update	0.973 px
Final reprojection RMSE	0.928 px
Median reprojection error	0.648 px
95th-percentile reprojection error	1.721 px
Maximum retained reprojection error	2.925 px
Camera baseline RMSE	0.00417 m
Positive-depth fraction	100%
BA convergence reported	Yes

The final reprojection RMSE is 0.928 px across 324 retained corner observations. All 22 objects are retained, although some image observations are excluded. With four corners per object and four images, the complete set would contain 352 corner observations.

The baseline RMSE is approximately 4.17 mm. This measures agreement with the sensor-derived distances used during optimization; it does not establish millimetre-level accuracy of the reconstructed objects. Similarly, the reprojection metrics describe agreement with the retained image measurements rather than independent ground-truth accuracy.

Table 2 gives the refined camera centres in the local coordinate system. The reference camera 0051_POS_073 remains at the origin.

Table 2: Refined camera centres relative to the reference camera of G0001. All coordinates are in metres.

Frame	C_x	C_y	C_z
0051_POS_073	0.000	0.000	0.000
0052_POS_105	20.994	-0.226	3.869
0053_POS_105	26.015	-0.368	4.693
0054_POS_113	32.412	-0.516	6.170

1.8 Image Overlay

Finally, we project the reconstructed corners back onto a selected image to compare them with the annotations. We also display the estimated object dimensions. This overlay provides a visual check of image alignment, while the geometry table provides the corresponding numerical evaluation.

Figure 1 shows the overlay for 0052_POS_105. For example, the reconstructed darkening object DHBW_OBJ_001 has an estimated width of 3.335 m and height of 0.264 m. These are reconstructed dimensions, not independently verified physical measurements.



Figure 1: Image overlay for local map G0001 on frame 0052_POS_105. The reconstructed object corners are projected onto the image and compared with the annotations. The displayed widths and heights are estimated from the reconstructed 3D corners.

2 Global Map Reconstruction

After reconstructing the local maps, we combine overlapping maps into one common coordinate system. We first estimate the rotation and translation between the maps, then combine their observations and refine the geometry using bundle adjustment. We also reconstruct missing objects when their image observations are supported by cameras already registered in the common map.

2.1 Aligning the Local Maps

Each local map uses its own reference camera. Therefore, the same object can have different coordinates in two local maps, even though both estimates describe the same physical object.

Suppose we want to align local map G_2 with the current global map G_1 . We estimate a rotation matrix $\mathbf{Q}$ and a translation vector $\mathbf{d}$ such that

$$\mathbf{X}^{G_1} = \mathbf{Q}\mathbf{X}^{G_2} + \mathbf{d}. \quad (11)$$

Here, $\mathbf{Q}$ aligns the coordinate axes, and $\mathbf{d}$ places the origin of G_2 in the coordinate system of G_1 . We use one transformation for the complete incoming map. No additional scale factor is estimated because the local maps already have metric scale.

2.1.1 Alignment Using Shared Objects

First, we identify objects reconstructed in both maps using their object IDs. Their corresponding corners provide pairs of 3D points. We estimate $\mathbf{Q}$ and $\mathbf{d}$ by minimizing

$$\min_{\mathbf{Q}, \mathbf{d}} \sum_i \left\| \mathbf{x}_i^{G_1} - (\mathbf{Q}\mathbf{x}_i^{G_2} + \mathbf{d}) \right\|^2, \quad \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}, \quad \det(\mathbf{Q}) = 1. \quad (12)$$

We use singular value decomposition (SVD) to solve this rigid alignment problem. The reconstructed corners contain estimation errors, so one transformation will generally not make every pair coincide exactly. SVD finds the rotation that best aligns the selected corresponding points in the least-squares sense while preserving distances and excluding reflections.

After estimating the rotation, we obtain the translation from the point centroids:

$$\mathbf{d} = \bar{\mathbf{X}}^{G_1} - \mathbf{Q}\bar{\mathbf{X}}^{G_2}. \quad (13)$$

We also test subsets of shared objects to reduce the influence of inconsistent local estimates.

2.1.2 Alignment Using Shared Cameras

A camera appearing in both maps provides another estimate of the same transformation. For a shared camera f , we calculate

$$\mathbf{Q}_f = (\mathbf{R}_f^{G_1})^\top \mathbf{R}_f^{G_2}, \quad (14)$$

$$\mathbf{d}_f = \mathbf{C}_f^{G_1} - \mathbf{Q}_f \mathbf{C}_f^{G_2}, \quad (15)$$

where $\mathbf{R}_f^G$ rotates map coordinates into camera coordinates, and $\mathbf{C}_f^G$ is the camera centre expressed in that map.

Ideally, all shared cameras provide the same alignment. In practice, their estimates differ because the local maps were reconstructed independently. We

therefore treat the camera-based and object-based alignments as alternative proposals and test their geometric consistency. SVD is used to align the shared 3D corners; it does not average the camera-based rotations.

2.2 Combining the Geometry and Applying Bundle Adjustment

For a consistent alignment proposal, we bring the incoming object corners and camera poses into the common coordinate system:

$$\mathbf{X}_{\text{new}} = \mathbf{Q}\mathbf{X}_{\text{incoming}} + \mathbf{d}, \quad (16)$$

$$\mathbf{C}_{f,\text{new}} = \mathbf{Q}\mathbf{C}_{f,\text{incoming}} + \mathbf{d}, \quad (17)$$

$$\mathbf{R}_{f,\text{new}} = \mathbf{R}_{f,\text{incoming}}\mathbf{Q}^\top. \quad (18)$$

We then combine the image observations of shared objects and keep one estimate per object and camera. Repeated observations from overlapping local maps are counted only once.

Next, we apply bundle adjustment (BA) to refine the combined geometry. For a reconstructed corner $\mathbf{X}_j$, its predicted image position in camera f is

$$\hat{\mathbf{p}}_{fj} = \pi(\mathbf{K}\mathbf{R}_f(\mathbf{X}_j - \mathbf{C}_f)), \quad (19)$$

where $\mathbf{K}$ is the camera intrinsic matrix and $\pi([x, y, z]^\top) = [x/z, y/z]^\top$.

BA reduces the differences between these projected positions and the annotated, undistorted image points. We use a robust loss to limit the influence of large pixel errors and include sensor-derived camera baseline lengths to constrain metric scale.

During an individual merge, we refine the incoming objects and newly introduced cameras while keeping the existing cameras fixed. Global refinement then jointly adjusts the object corners and all camera poses except the reference pose.

We accept the update only when the reprojection errors, viewing geometry, and baseline consistency satisfy the quality checks without removing previously accepted objects. Otherwise, we retain the current map and defer the incoming local map.

2.3 Reconstructing Missing Objects

Some objects have image annotations but no accepted 3D estimate contributed by the local maps. Once their observing cameras are registered in the common coordinate system, we can attempt to reconstruct these objects directly.

For example, suppose a door is observed in frame F_1 of one local map and frame F_4 of another. Neither local map individually contains enough usable views to reconstruct the door. After aligning the maps through other shared objects, the poses of F_1 and F_4 are available in the same coordinate system.

We then use the door’s original corner annotations and the registered camera projection matrices,

$$\mathbf{P}_f = \mathbf{K} [\mathbf{R}_f \quad -\mathbf{R}_f \mathbf{C}_f], \quad (20)$$

to triangulate its corners. The resulting points are already expressed in the common map because the camera poses use that coordinate system. We do not apply $\mathbf{Q}$ and $\mathbf{d}$ directly to image pixels.

We first refine the new corners with the camera poses fixed, then perform joint BA and quality checks before accepting the update. Recovery requires sufficient usable views and reliable viewing geometry; the presence of annotations alone does not guarantee successful reconstruction.

2.4 DHBW Global Map Results

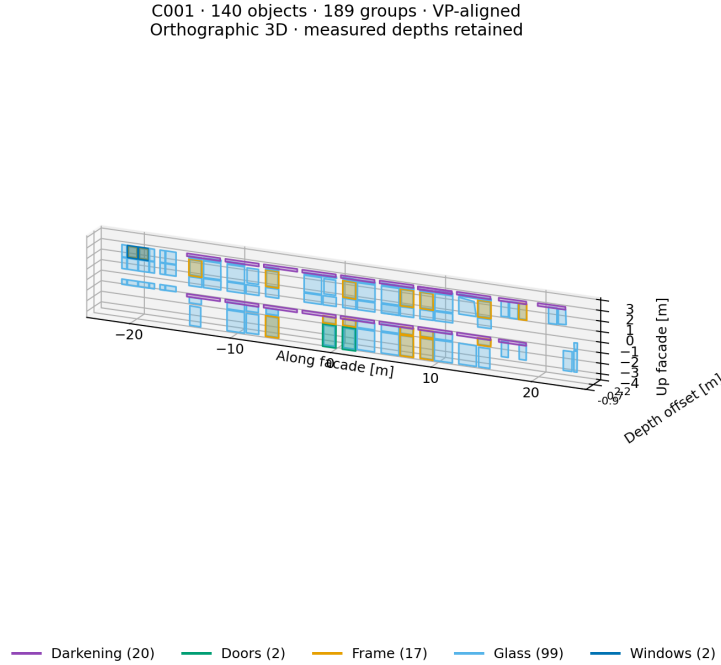


Figure 2: Orthographic 3D view of the DHBW global map C001, containing 140 reconstructed objects. The display is aligned using vanishing-point directions, with reconstructed depth retained.

The selected global map, C001, combines 189 local groups and 29 unique camera frames. It contains 140 objects and 560 reconstructed corners, supported by 2,312 retained corner observations. Its reference frame is 0051_POS_073.

The reconstructed objects comprise 99 glass objects, 20 darkening objects, 17 frames, two windows, and two doors. Recovery is recorded for 59 objects.

This flag describes reconstruction history: some recovered objects also gained contributions from local maps during subsequent merging.

Table 3: Reconstruction results for the DHBW global map C001.

Metric	Result
Contributing local groups	189
Registered camera frames	29
Reconstructed objects	140
Reconstructed corners	560
Used corner observations	2,312
Reprojection RMSE	1.105 px
Median reprojection error	0.833 px
95th-percentile reprojection error	2.094 px
Maximum retained reprojection error	2.996 px
Minimum ray-angle metric	6.501°
Positive depth in supporting cameras	Yes
Camera baseline RMSE	0.01452 m
Camera-position RMSE relative to sensor poses	2.580 m

The reprojection RMSE shows that the reconstructed geometry agrees with the retained image measurements to approximately one pixel. These errors are calculated after BA and observation screening, so they measure internal consistency rather than independent reconstruction accuracy.

The baseline RMSE measures agreement with sensor-derived distances already used during optimization. It therefore does not establish centimetre-level object accuracy. The camera-position RMSE is calculated after the provisional world transformation inherited from the reference local map; it is not a direct measurement of object-position error.

For visualization, we use horizontal and vertical vanishing-point directions to orient the map along the facade. The 3D view retains the reconstructed depth, while the front view omits depth for display. This display transformation does not change the saved geometry or estimate wall positions.

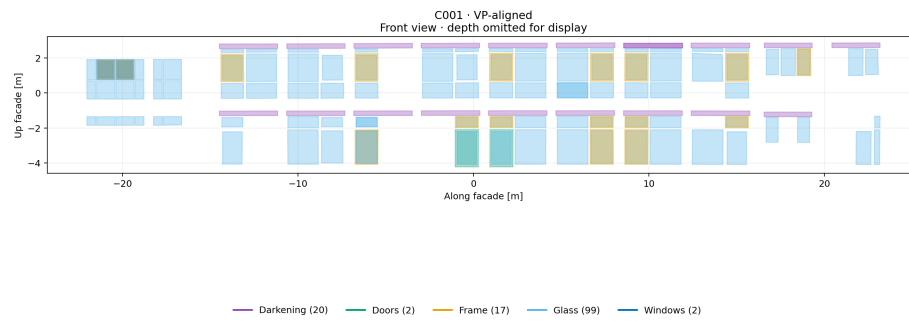


Figure 3: Front view of the DHBW global map C001. Depth is omitted for visualization, showing the arrangement of reconstructed objects along the facade.